

Protocol and Architecture

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How the system is assembled, and why each boundary exists

Asher Wilson, Founder

asher@maschine.dev

Abstract

A technical description of Maschina for readers who intend to check the claims rather than accept them. Six services, each with one job and a boundary enforced by continuous integration rather than by convention. An append-only record fenced by lease epoch. A signing path in which the party that decides never holds a key. Operational status is stated in document 14.

1. Shape

Maschina is six services and a set of shared libraries. Every boundary described below is checked on every commit, and a change that crosses one fails the build before a human sees it.

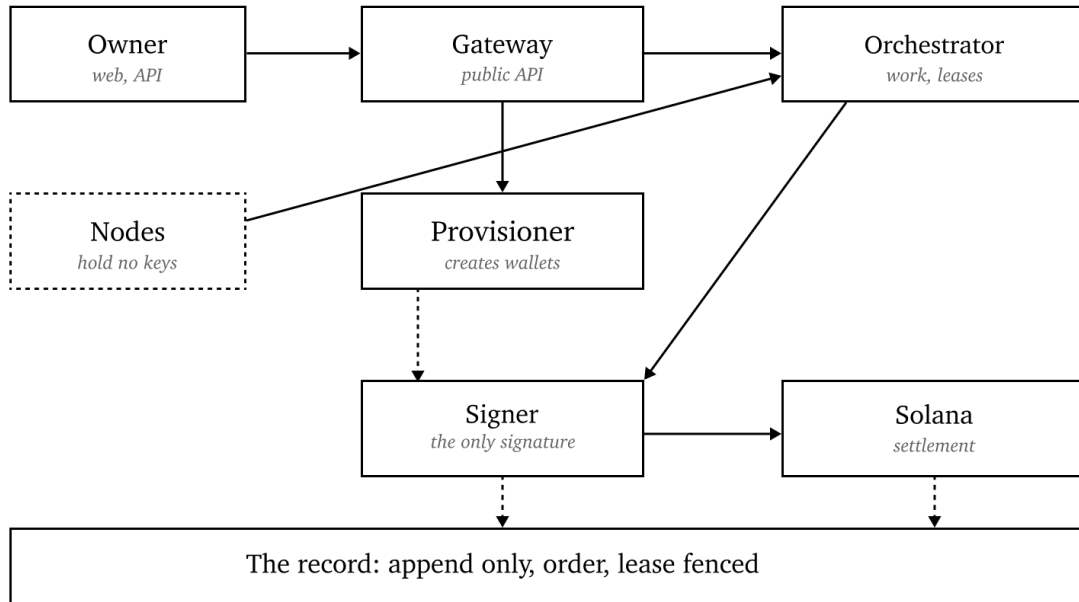


Figure 1. The services and their boundaries. The dashed path is untrusted: nodes decide and propose, and hold nothing.

The gateway is the public interface and the only service the application speaks to. The orchestrator owns work and leases. The signer is the only service that can obtain a signature. The provisioner is the only service that can create a wallet, and it cannot sign. Nodes run machines, dial outward, and never touch the database. Chain libraries exist in exactly one package.

2. Why the boundaries sit where they do

Each one removes a category of failure.

Nodes hold no keys, which is what makes a permissionless network safe rather than merely aspirational. An untrusted node can propose something and have it refused, or it can stop working and lose its lease. That is the entire network available to it.

The provisioner creates wallets and cannot sign; the signer signs and cannot create wallets or alter policies. The two use distinct provider credentials with distinct permissions, so the compromise of either is survivable and neither is sufficient.

The gateway resolves the owner from the session and asks the database for “this owner’s machine with this identifier” in a single statement. Someone else’s machine is not found, rather than refused with a message confirming it exists.

Chain libraries live in one package, so an additional venue is an adapter, and a second chain, or a deliberately fake chain used in tests, is a package rather than a rewrite.

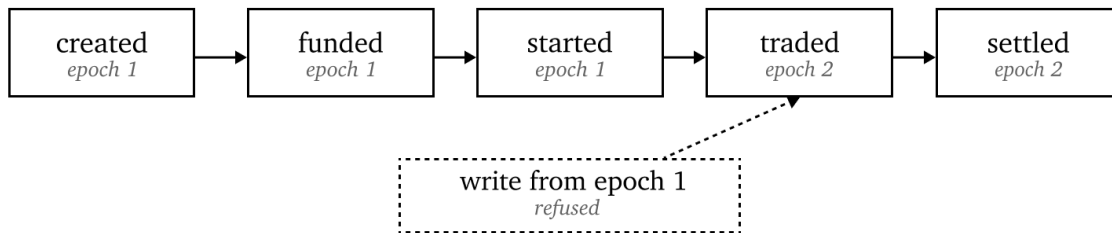
3. The record

State is derived, never stored. Reading a machine's events in order produces its state, its budget and its limits.

Two invariants are enforced at the one function that writes, because an invariant enforced in several places is eventually enforced in none.

Schema conformance. Every event is validated against the schema for its type. An append-only record cannot be corrected later, so an event written incorrectly is permanent.

Lease fencing. A write carries the epoch of the lease its author holds, and the comparison happens inside the same statement as the insert.



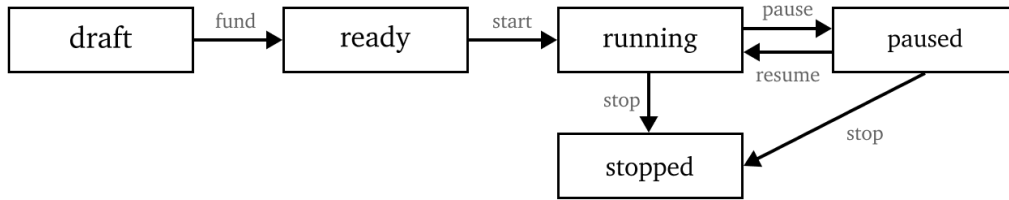
The epoch check and the insert are one statement, so two writers cannot both win.

Figure 2. The record is append only and fenced. A writer from a superseded epoch is refused by the same statement that would have accepted it.

Ordering is by acceptance time, then by identifier, and. Identifiers are time ordered so they never tie.

4. Lifecycle

Reading history is deliberately forgiving: an unusual sequence still produces an answer, because refusing to interpret the past is not an option for a system that must start up and continue. Accepting a change is strict: a transition is validated before anything is written, so the record only ever gains changes that were permitted at the moment they occurred.



Stopped is terminal. A machine that could restart itself is one you cannot switch off.

Figure 3. The lifecycle. Every transition that exists is checked before it is recorded, and the one that does not exist is the one that would undo a stop.

5. One action, end to end

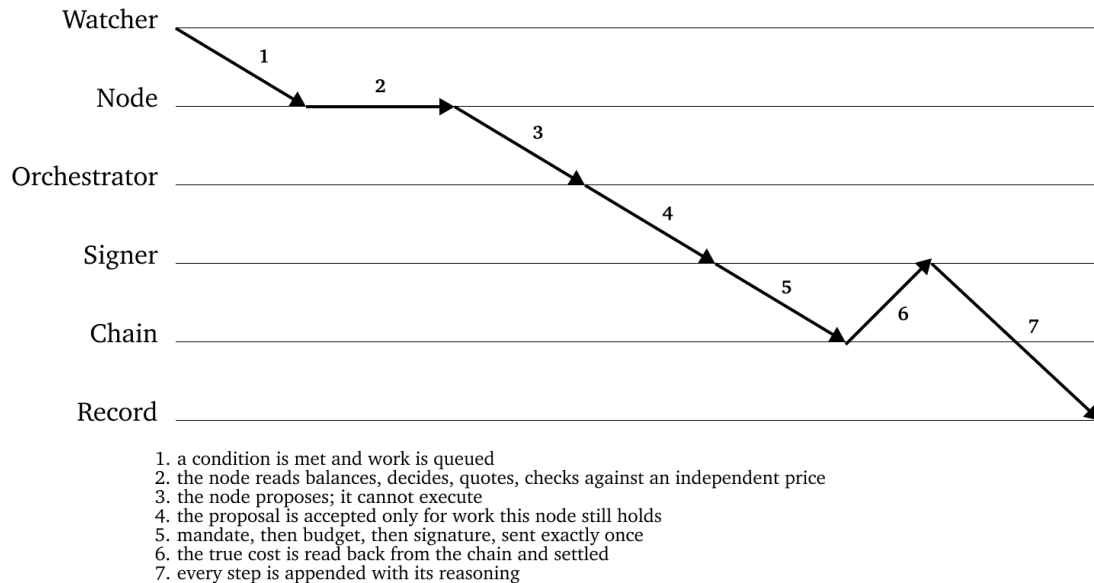


Figure 4. The path of a single action across services.

The watcher observe a condition and queues work, once per crossing, with a hysteresis band so a price resting on a level does not fire repeatedly and a minimum interval between runs.

A node claims the work, takes a lease and renews it. It reads balances, evaluates the machine's decision function, obtains a quote, checks the quote against an independent price source, and proposes.

The orchestrator forwards the proposal only for work that node still holds. The signer applies the mandate, reserves budget under a row lock, signs through a provider, and sends exactly once. Retries reuse the same signature.

The true cost is read back from the chain, counting native and wrapped balances together and removing fees, and the budget is settled against that number rather than the estimate. Every step is appended with its reasoning.

6. Machine kinds

A machine kind is a deterministic decision function: settings, balances and history in, a proposed action or nothing out.

It cannot execute, hold a key or contact a venue. Determinism is a security property rather than an aesthetic preference, because a decision that can be recomputed from its inputs can be replayed and checked by a stranger, which is what makes it possible to pay strangers to run machines they did not write.

7. Execution quality

Two numbers are recorded for every action: time from decision to landing, and the gap between the quoted and the realized cost.

Both are published per machine. They are the difference between a strategy that works on paper and one that works, and they are the metric the network exists to improve.

8. Interfaces

The public interface is described by a document generated from the route definitions rather than maintained beside them, so it cannot drift.

Amounts are strings of digits in base units, never floating point. Times are ISO 8601 in UTC. An unset limit is absent rather than zero, because zero and unset mean opposite things. Errors carry a stable code, a human readable message and a request identifier.

9. Operations

Services run as containers. The host exposes no ports to the internet and the public service is reachable only through an outbound tunnel, so nothing is listening and no certificate is ever renewed on the machine. The database is unreachable from outside the host. Backups are encrypted with a key held elsewhere, taken nightly, and the restore procedure is performed rather than assumed.

Releases build images pinned by digest, each carrying signed provenance and software bill of materials, and deployment happens only after those images exist.

10. Testing

Every behavior has a test that failed before the behavior existed. Integration tests run against real PostgreSQL and real chain endpoints rather than mocks, because the failures worth catching are the ones mocks cannot produce: concurrency, ordering, partial failure, and the gap between an estimate and reality.

Concurrency is proved rather than argued. Twenty actions racing for a budget that fits ten result in exactly ten.